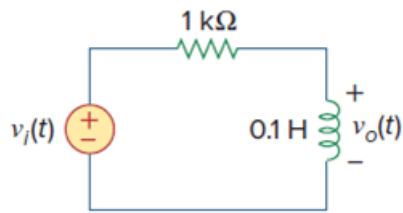


Problem

Determine what type of filter is in Fig. 14.87. Calculate the corner frequency f_c .

Figure 14.87:



Step-by-step solution

Step 1 of 5

Refer to Figure 14.87 in the textbook.

In frequency domain, the inductor is represented as $j\omega L$.

Draw the circuit in frequency domain.

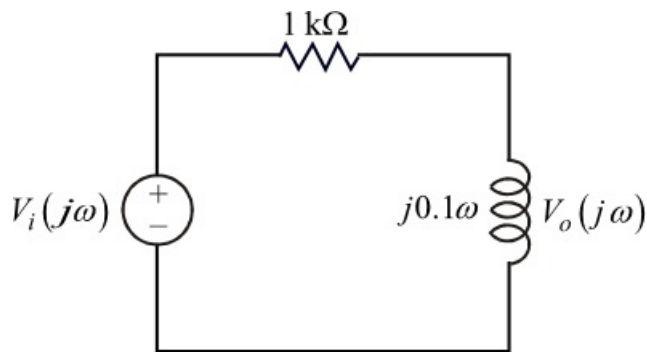


Figure 1

Step 2 of 5

Write the expression for output voltage.

$$\mathbf{V}_o(j\omega) = \mathbf{V}_i(j\omega) \left[\frac{0.1j\omega}{1000 + 0.1j\omega} \right]$$

$$\frac{\mathbf{V}_o(j\omega)}{\mathbf{V}_i(j\omega)} = \left[\frac{0.1j\omega}{1000 + 0.1j\omega} \right]$$

$$H(j\omega) = \left[\frac{0.1j\omega}{1000 + 0.1j\omega} \right]$$

Calculate the magnitude of the function.

$$|H(j\omega)| = \frac{0.1\omega}{\sqrt{1000^2 + (0.1\omega)^2}}$$

Step 3 of 5

Calculate the magnitude at $\omega = 0$.

$$\begin{aligned} |H(0)| &= \frac{0.1(0)}{\sqrt{1000^2 + (0.1(0))^2}} \\ &= 0 \end{aligned}$$

Calculate the magnitude at $\omega = \infty$.

$$\begin{aligned} |H(\infty)| &= \frac{0.1(\infty)}{\sqrt{1000^2 + (0.1(\infty))^2}} \\ &= 1 \end{aligned}$$

From the calculations, it is clear that, the filter circuit allows only high frequencies components and attenuates the low frequencies components.

Therefore, the circuit is high pass filter.

Step 4 of 5

Consider the transfer function.

$$\begin{aligned} H(j\omega) &= \left[\frac{0.1j\omega}{1000 + 0.1j\omega} \right] \\ &= \left[\frac{0.1j\omega}{1000 \left(1 + j \frac{0.1\omega}{1000} \right)} \right] \\ &= \frac{j \frac{\omega}{10,000}}{\left(1 + j \frac{\omega}{10,000} \right)} \end{aligned}$$

From the transfer function, the value of $\omega_c = 10,000$ rad/s.

Step 5 of 5

Calculate the value of the corner frequency, f_c .

$$f_c = \frac{\omega_c}{2\pi}$$

Substitute 10,000 rad/s for ω_c .

$$\begin{aligned} f_c &= \frac{10,000}{2\pi} \\ &= 1591.55 \text{ Hz} \end{aligned}$$

Thus, the corner frequency of the circuit is 1591.55 Hz.